The ultraviolet part of the $C_1 \otimes B_2$ row

The $z \to x$ divergence, its coefficient, and nothing else

Including the factor–of–2 audit of the $S_1 = (P\cdot r)(s\cdot r)/r^4$ term

0. Answer to the question first

No factor of 2 is missing. The coefficient of S_1 in the bracket is $c_1 = d_\perp \xi\bar\xi = 2\xi\bar\xi$, and the angular average supplies a compensating $\frac{1}{2}$:

$$\Big\langle \frac{(P\cdot r)(s\cdot r)}{r^4} \Big\rangle = \frac{s\cdot P}{d\,r^2} \xrightarrow{d=2} \frac{s\cdot P}{2r^2} \qquad \Longrightarrow \qquad c_1 \Big\langle S_1 \Big\rangle = 2\xi\bar\xi \cdot \frac{s\cdot P}{2r^2} = \xi\bar\xi\,\frac{s\cdot P}{r^2},$$

which is the $\xi(1-\xi)$ of P_{gg} with coefficient exactly 1, as it must be. Dropping the $\frac{1}{2}$ would give $2\xi(1-\xi)$ and the wrong splitting function; keeping the 2 of c_1 and the $\frac{1}{2}$ of the average is what makes it come out right.

There *is* a place where a spurious 2 can appear, and it is worth stating explicitly because the earlier notes were sloppy about it: $d_\perp$. The instantaneous δ^{im} vertex carries an explicit $d_\perp$, and the angular average carries $1/d$. These are the same dimension, so they cancel *exactly*:

$$c_1 \big\langle S_1 \big\rangle = \frac{d_\perp}{d}\,\xi\bar\xi\,\frac{s\cdot P}{r^2} = \xi\bar\xi\,\frac{s\cdot P}{r^2} \qquad \text{for } d = d_\perp, \text{ any } d_\perp.$$

Writing $\frac{d_\perp}{2}\xi\bar\xi$ — i.e. keeping $d_\perp$ in the vertex but averaging in exactly two dimensions — is the inconsistent hybrid. It agrees at $d_\perp = 2$, so nothing computed so far changes, but it wrongly suggests that the $\xi(1-\xi)$ piece of P_{gg} receives an $O(\epsilon)$ correction from this term. It does not.

1. Where the divergence is

With $r = z - x$, $s = y - x$, $P = x' - w'$, $\xi = p^+/k^+$, $\bar\xi = 1 - \xi$, and

$$p^+(y-x)^2 + (k^+ - p^+)(y-z)^2 = k^+\bar\xi D, \qquad D = (r-s)^2 + M^2, \qquad M^2 = \frac{\xi}{\bar\xi}s^2,$$

the transverse integrand of Eq. (1.1) is singular only at $r \to 0$, i.e. $z \to x$. There $D \to D_0 \equiv s^2 + M^2 = s^2/\bar\xi$ and $(r-s)^2 \to s^2$, both finite and non–zero, and $e^{iq\cdot r} \to 1$. So the whole question is the small–r behaviour of the bracket.

In two dimensions $d^2r = r\,dr\,d\theta$, so a term going like r^{-n} is logarithmically divergent for $n = 2$ and convergent for $n \le 1$. The bracket contains only $1/r^2$ and $1/r^4$ denominators, so only two kinds of term can diverge:

$$S_1 = \frac{(P\cdot r)(s\cdot r)}{r^4} \qquad \text{and} \qquad S_2 = \frac{s\cdot P}{r^2}.$$

Everything else is either odd in r at leading order (the $\frac{P\cdot r}{r^2} \sim 1/|r|$ structures, integrable), or bounded, or carries a numerator that vanishes at the point where its denominator does. In particular the $(s-r)^2$ structures behave as $1/|r-s|$ near $r = s$ and are integrable there, so $z \to y$ produces no divergence.

2. The angular average — where the 2 lives

In d transverse dimensions $\langle \hat r^{\,i} \rangle = 0$ and $\langle \hat r^{\,i}\hat r^{\,j} \rangle = \delta^{ij}/d$, so

$$\Big\langle \frac{(P\cdot r)(s\cdot r)}{r^4} \Big\rangle = P_i s_j \frac{\delta^{ij}}{d\,r^2} = \frac{s\cdot P}{d\,r^2}, \qquad \Big\langle \frac{s\cdot P}{r^2} \Big\rangle = \frac{s\cdot P}{r^2}, \qquad \Big\langle \frac{P\cdot r}{r^2} \Big\rangle = 0.$$

Numerically, $\big\langle \frac{(P\cdot r)(s\cdot r)}{r^4} \big\rangle \frac{r^2}{s\cdot P} = 0.500000000000$ at $d = 2$ (§6A). The three facts to keep straight:

object	value	where it comes from
coefficient of S_1 in the bracket	$c_1 = d_\perp\,\xi\bar\xi$	Θ_1 gives $d_\perp\xi\bar\xi$; $\Theta_3,\Theta_5,\Theta_6$ give $-2\bar\xi-2\xi+2=0$
angular average of S_1	$\dfrac{s\cdot P}{d\,r^2}$	$\langle\hat r^i\hat r^j\rangle=\delta^{ij}/d$
product	$\dfrac{d_\perp}{d}\xi\bar\xi\dfrac{s\cdot P}{r^2}=\xi\bar\xi\dfrac{s\cdot P}{r^2}$	the $\xi(1-\xi)$ of P_{gg}

So the 2 in $c_1=2\xi\bar\xi$ and the $\frac12$ in $\langle S_1\rangle$ are two sides of the same $d_\perp=2$, and they cancel. Any answer in which $\xi(1-\xi)$ appears with coefficient 2 instead of 1 has used one of them and not the other.

3. Term by term

Write the bracket of Eq. (1.1) as $\sum_i w_i(\xi)V_i$ with $w=(\xi\bar\xi,\ \frac{\xi}{\bar\xi},\ \bar\xi,\ \frac{\bar\xi}{\xi},\ \xi,\ 1)$ and V_i the r–structures of your (1.4)–(1.10). Taking $r\to0$ and averaging over the angle:

term	structure that survives	$V_i\big\vert_{\mathrm{UV}}$ in units of $\frac{s\cdot P}{P^2r^2}$	$\times\,w_i$
Θ_1 (instantaneous $\delta^{im}d_\perp$)	$\frac{d_\perp}{2}\cdot\frac{2(P\cdot r)(s\cdot r)}{r^4}$	$\dfrac{d_\perp}{d}=+1$	$+\xi\bar\xi$
Θ_2	$\frac{s\cdot P}{r^2}$	$+1$	$+\dfrac{\xi}{\bar\xi}$
Θ_3	$-\frac{2(P\cdot r)(s\cdot r)}{r^4}$	$-\dfrac{2}{d}=-1$	$-\bar\xi$
Θ_4	$\frac{s\cdot P}{r^2}$	$+1$	$+\dfrac{\bar\xi}{\xi}$
Θ_5	$-\frac{2(P\cdot r)(s\cdot r)}{r^4}$	$-\dfrac{2}{d}=-1$	$-\xi$
Θ_6	$+\frac{2(P\cdot r)(s\cdot r)}{r^4}$	$+\dfrac{2}{d}=+1$	$+1$

The last three cancel identically, $-\frac{2}{d}(\bar\xi+\xi-1)=0$, in *any* d. What is left is

$$\boxed{\begin{gathered}\text{bracket}\ \xrightarrow[r\to0]{}\ C_{\mathrm{UV}}(\xi)\,\frac{s\cdot P}{P^2r^2},\qquad C_{\mathrm{UV}}(\xi)=\xi\bar\xi+\frac{\xi}{\bar\xi}+\frac{\bar\xi}{\xi}=\xi(1-\xi)-2+\frac{1}{\xi(1-\xi)}=\frac{P_{gg}(\xi)}{2N_c},\\ P_{gg}(\xi)=2N_c\Big[\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}+\xi(1-\xi)\Big].\end{gathered}}$$

The correspondence is one to one: $\xi(1-\xi)$ from the instantaneous Θ_1, $\frac{\xi}{1-\xi}$ from Θ_2, $\frac{1-\xi}{\xi}$ from Θ_4; $\Theta_3,\Theta_5,\Theta_6$ contribute nothing to the ultraviolet. Checked directly against the tensors of (1.1) — at $\xi=0.3$ the six coefficients come out $(0.21,\ 0.4286,\ -0.7,\ 2.3333,\ -0.3,\ 1)$, i.e. $(\xi\bar\xi,\ \frac{\xi}{\bar\xi},\ -\bar\xi,\ \frac{\bar\xi}{\xi},\ -\xi,\ 1)$ to seven digits (§6B).

4. From an asymptotic statement to an exact one

The angular argument of §2–§3 identifies the divergence but only as $r\to0$. It becomes an exact identity by splitting S_1 into its trace and traceless parts,

$$\frac{(P\cdot r)(s\cdot r)}{r^4}=\underbrace{\frac{s\cdot P}{2r^2}}_{\text{trace}}+\underbrace{P_is_j\Big[\frac{r^ir^j}{r^4}-\frac{\delta^{ij}}{2r^2}\Big]}_{\text{traceless}},$$

the traceless bracket being orthogonal to every function of r^2 alone and hence contributing no logarithm. Since the coefficient of S_1 is $2\xi\bar\xi$ and that of S_2 is $\frac{\xi}{\bar\xi}+\frac{\bar\xi}{\xi}$, and $\frac12(2\xi\bar\xi)+\frac{\xi}{\bar\xi}+\frac{\bar\xi}{\xi}=C_{\mathrm{UV}}$,

$$\boxed{P^2\times\text{bracket}\ =\ C_{\mathrm{UV}}(\xi)\,\frac{s\cdot P}{r^2}\ +\ \big(\text{integrable at }r=0\big),\qquad\text{for every }r,}$$

so the ultraviolet content of the row is one term, and it is unambiguous. (The remainder is written out in the companion note; it is not needed here.)

5. The z integral of the UV term, and the p^+ integral

Regulate $r^2 \to r^2 + m^2$. Only the UV term needs it:

$$\int d^2r \, \frac{e^{iq\cdot r}}{(r^2+m^2)\,D} = \frac{2\pi}{D_0} K_0(m|q|) + \text{finite} = \frac{\pi}{D_0} \log \frac{D_0}{m^2} + \text{finite}, \qquad D_0 = \frac{s^2}{\bar{\xi}}.$$

Inserting this in (1.11), the explicit $\bar{\xi}$ of $1/D_0 = \bar{\xi}/s^2$ cancels the $1/\bar{\xi}$ of the longitudinal denominator exactly, and the transverse structure collapses to a product of two Weizsäcker–Williams kernels:

$$\boxed{\frac{1}{\bar{\xi}} \frac{1}{P^2} \Big[\text{UV term}\Big] = \pi \, \frac{P_{gg}(\xi)}{2N_c} \, \frac{(x'-w')\cdot(y-x)}{(x'-w')^2\,(y-x)^2} \, \log \frac{(y-x)^2}{\bar{\xi}\, m^2}.}$$

Now the p^+ integral, $dp^+ = k^+ d\xi$, $\ell_k = \log(k^+/\Lambda)$, $L = \log\big[(y-x)^2/m^2\big]$; the logarithm above is $L + \log \frac{1}{1-\xi}$. All six moments are elementary:

piece of $C_{\text{UV}} = P_{gg}/2N_c$	from	$\int_\lambda^{1-\lambda} d\xi$	$\int_\lambda^{1-\lambda} d\xi \, \log\frac{1}{1-\xi}$
$\xi(1-\xi)$	Θ_1, instantaneous	$\frac{1}{6}$	$\frac{5}{36}$
$\dfrac{\xi}{1-\xi}$	Θ_2	$\ell_k - 1$	$\frac{1}{2}\ell_k^2 - 1$
$\dfrac{1-\xi}{\xi}$	Θ_4	$\ell_k - 1$	$\frac{\pi^2}{6} - 1$
total		$2\ell_k - \frac{11}{6}$	$\frac{1}{2}\ell_k^2 + \frac{\pi^2}{6} - \frac{67}{36}$

$$\int_\Lambda^{k^+-\Lambda} \frac{dp^+}{2\pi} \, \frac{P_{gg}(\xi)}{2N_c} \, \log \frac{k^+(y-x)^2}{(k^+-p^+)m^2} = \frac{k^+}{2\pi} \left[\left(2\ell_k - \frac{11}{6}\right) \log \frac{(y-x)^2}{m^2} + \frac{\ell_k^2}{2} + \frac{\pi^2}{6} - \frac{67}{36} \right].$$

Three readings. The double logarithm comes only from Θ_2 and the $\pi^2/6$ only from Θ_4; it is the instantaneous Θ_1 that turns -2 into $-\frac{11}{6}$ and -2 into $-\frac{67}{36}$. And

$$-\frac{11}{6} \times 2N_c = -\frac{11N_c}{3} = -b\Big|_{N_f=0},$$

so this row supplies, on its own, the $\frac{11N_c}{6}\delta(1-\xi)$ of the regularised splitting function — a running–coupling contribution.

> **The constant, in the paper's normalisation.** $C_{\text{UV}} = \xi(1-\xi) - 2 + \frac{1}{\xi(1-\xi)}$ is literally the weight of Eq. (C.10) of arXiv:1610.03453. Because C_{UV} is symmetric under $\xi \leftrightarrow 1-\xi$, $\int C_{\text{UV}} \log[\xi(1-\xi)] = 2\int C_{\text{UV}} \log(1-\xi)$, so
>
> $$\int_\lambda^{1-\lambda} d\xi \, C_{\text{UV}} \log\big[\xi(1-\xi)\big] = -\ell_k^2 + \frac{67}{18} - \frac{\pi^2}{3} = -2\left(\frac{\ell_k^2}{2} + \frac{\pi^2}{6} - \frac{67}{36}\right),$$
>
> which is (C.10) exactly. That $\frac{67}{18} - \frac{\pi^2}{3}$ is the source of the $\left(\frac{67}{9} - \frac{\pi^2}{3}\right)N_c$ of K_{JSJ}, Eqs. (2.64), (2.66), (4.36).

6. Dimensional regularisation

The entire regulator dependence is the single factor $\pi \log(1/m^2)$. Matching against $\int d^{d_\perp}r \, e^{iq\cdot r}/r^2 = \pi^{1+\epsilon} 2^{2\epsilon} \Gamma(\epsilon)(q^2)^{-\epsilon}$ with $d_\perp = 2 + 2\epsilon_{\text{UV}}$ gives

$$\log \frac{1}{m^2} \;\longleftrightarrow\; \frac{1}{\epsilon_{\text{UV}}} + \log\big(\pi e^{\gamma_E} \mu^2\big),$$

i.e. the $1/\epsilon_{\text{UV}}$ that cancels against the $-2/\epsilon$ of the gluon self–energy counterterm. (With $d_\perp = 2 - 2\epsilon$ the pole is $-1/\epsilon$.) As shown in §0 and §2, the $\xi(1-\xi)$ coefficient is $d_\perp/d = 1$ exactly, so continuing the transverse plane produces **no** $O(\epsilon)$ correction to it — the $d_\perp$ of the instantaneous vertex and the $1/d$ of the angular average cancel. (This corrects a remark in the first companion note, which used the hybrid $\frac{d_\perp}{2}\xi\bar{\xi}$ and suggested such a correction existed. Nothing computed at $d_\perp = 2$ changes.)

7. Checks

Every number below is produced by `zint/src/run_uv.py`. Block B builds the six structures directly from the tensors of Eq. (1.1) — not from the reduced forms — and averages them numerically over the angle; block D measures the coefficient of $\log(1/m^2)$ on the full two–dimensional integral with a regulator applied to the original integrand, so it is independent of every algebraic step above.

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kinematics: s = [ 0.7 -0.4]  P = [1.3 0.5]   s^2 = 0.6500  s.P = 0.7100

A. the angular average, exactly
   < (P.r)(s.r)/r^4 > r^2/(s.P) = 0.500000000000     1/d = 1/2 = 0.5
   < (P.r)/r^2 >     r  /|P|    = -4.012e-17      (odd, vanishes)

B. UV coefficient of each Theta_i, from the raw tensors of Eq. (1.1)
   xi=0.30 : Theta_i UV coefficients [ 0.21       0.4285714 -0.7        2.3333333 -0.3        1.       ]
               = [0.21     0.42857  0.7      2.33333  0.3      1.      ]  (weights)  ->  sum 2.971904779   C_UV =
2.971904762
   xi=0.65 : Theta_i UV coefficients [ 0.2275     1.8571429 -0.35       0.5384615 -0.65       1.       ]
               = [0.2275   1.85714  0.35     0.53846  0.65     1.      ]  (weights)  ->  sum 2.623104405   C_UV =
2.623104396

C. the cancellation among Theta_3, Theta_5, Theta_6 holds in any d
   sum = (-d*xi**2 - d*(xi - 1)**2 - d_perp*xi**2*(xi - 1)**2)/(d*xi*(xi - 1))
   at d = d_perp : (-xi**2*(xi - 1)**2 - xi**2 - (xi - 1)**2)/(xi*(xi - 1))
   Pgg/(2Nc)     : (-xi**2*(xi - 1)**2 - xi**2 - (xi - 1)**2)/(xi*(xi - 1))
   identical?    : True

D. the log coefficient measured on the FULL 2D integral (independent of all the algebra)
   d W / d log(1/m^2) of  int d^2r e^{iq.r} bracket/D , r^2 -> r^2+m^2 :
      xi=0.30  measured +7.137881   predicted C_UV (s.P) pi/D_0 = +7.138842   rel 1.3e-04
      xi=0.65  measured +3.150126   predicted C_UV (s.P) pi/D_0 = +3.150493   rel 1.2e-04
   (from RESULTS.txt, block D; regulator r^2 -> r^2+m^2 applied to the original integrand)

E. the p+ integrals of C_UV = Pgg/(2Nc)   (mpmath, 30 digits, lambda = 1e-9)
   int C_UV                 = 39.613198342559    2 l_k - 11/6            = 39.613198340559
   int C_UV log(1/(1-xi))   = 214.51069645084    l_k^2/2 + pi^2/6 - 67/36 = 214.51069643011
   int C_UV log(xi(1-xi))   = -429.02139290167   -l_k^2 + 67/18 - pi^2/3   = -429.02139286022   [= paper (C.10)]
   the three pieces separately:
      int xi(1-xi)  [Theta_1] = 0.166666666667   predicted 0.166666666667
      int xi/(1-xi) [Theta_2] = 19.7232658379   predicted 19.7232658369
      int (1-xi)/xi [Theta_4] = 19.7232658379   predicted 19.7232658369
      int xi(1-xi)  [Theta_1] log(1/(1-xi)) = 0.138888888889   predicted 0.138888888889
      int xi/(1-xi) [Theta_2] log(1/(1-xi)) = 213.726873496   predicted 213.726873474
      int (1-xi)/xi [Theta_4] log(1/(1-xi)) = 0.644934065848   predicted 0.644934066848
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